

Ong Zheng Kai Ethan

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EDUCATION

Nanyang Technological University

Aug. 2021 – Jun. 2025

Bachelor of Engineering (Honours), Mechanical Engineering

Singapore

- **GPA: 4.56/5.0; First Class Honours**
- Specialization: Robotics and Mechatronics
- **Industry-Sponsored Bachelor Thesis in Collaboration with HMGICS:** 3D Gaussian Splatting Techniques in Reconstructing Car Components for Automotive Assembly
- Relevant Courses: Intro to Data Science and Artificial Intelligence, Measurements and Sensing Systems, Mechatronics Systems Interfacing, Mechatronics Engineering Design

Technische Universität München

Apr. 2024 – Sep. 2024

Semester Exchange, Mechanical Engineering

Munich, Germany

- **GPA: 2.5/5.0; Gut (Good)**
- Relevant Masters-level Courses: Computational Thermo-Fluid Dynamics, Cyber Physical Systems
- Relevant Bachelor-level Courses: Automatic Control Engineering, Mobile Computer Vision

WORK EXPERIENCE

Panasonic R&D Center Singapore

Jan. 2025 – Present

Research and Development Intern

Singapore

- Improved the realism of 3D indoor reconstruction via SLAM, point cloud editing, and radiance field techniques
- Reduced rendering times by 200% by introducing asynchronous loading and simplifying fragment shader logic
- Containerized experimental features for easy redeployment and installation on multiple linux servers
- Connected assets database to user-friendly GUI for users to add and edit dynamic lighting and furniture meshes
 - **Stack:** [React.js](#), [Three.js](#), [Node.js](#), Axios, MongoDB, Docker, GLSL, WebGL, Javascript

Mecatron

Oct. 2024 – Present

Software Lead

Singapore

- Champions, Singapore Autonomous Underwater Vehicle Competition 2025, besting 40 teams from 20 countries
- Managing 10 software engineers by defining project lifecycles, conducting sprints, ensuring documentation etc.
- Created action servers and pub/sub frameworks in ROS, planned competition-winning missions using Behavior Trees. Configured Pixhawk Controllers for multi-servo and thruster setups, including SITL testing in UnitySim
- Streamlined training and recruitment via a computer vision screening task and a comprehensive newcomer tutorial which teaches simulation, state machine, and version control concepts specific to our workflow
 - **Stack:** Git, Notion, C++, Python, Unity, ROS, Docker, Linux, OpenCV

Hyundai Motor Group Innovation Center Singapore

Aug. 2024 – May 2025

Student Researcher

Singapore

- Improved digital reconstruction of car parts using 3D Gaussian Splatting for vision-based motion-planning
- Reduced image dataset by 65% by optimising robotic arm trajectories, multi-view stereo depth and normal estimation, and image transforms. Improved point cloud density and cleanliness and camera pose accuracy
- Constructed data capture, background masking, rasterization, 3D segmentation and pose estimation pipeline
 - **Stack:** ABB RAPID, Python, COLMAP, CloudCompare, MeshLab, Open3D, Photoneo PhoXi

Fling Asia

Jun. 2023 – Apr. 2024

Software Engineer | Dec. 2023 – Apr. 2024

Bangkok, Thailand

- Directed marketing initiatives, product demonstrations, and inventory audits for Fling's drone operations and software across 10+ warehouses, securing ongoing stock-checking contracts with 90% of them
- Led development of a cross-platform bulk labelling program in Qt, supporting Zebra printing and QR encoding, successfully deployed in Ceva Logistics Shell and DHL Harley-Davidson, Thailand

Engineering Intern | Jun. 2023 – Nov. 2023

- Designed, 3D printed, & simulated FEA on drone-lighting add-ons to assist visual navigation in dim conditions
- Improved ML inference speeds on video streams by 300% using multi-processing, tensors & GPU acceleration
- Developed internal tools which abstracted SQL scripts, database editing, and core CLI commands into a GUI
 - **Stack:** PySide6 (Qt), Python, C++, SQL, PyTorch, Fusion360, DJI, MariaDB, Google Cloud

Republic of Singapore Air Force

Oct. 2018 – Jul. 2021

Weapon Systems Officer (Fighter) Trainee

Singapore

- Appointed as Commander Air Training Group, coordinating flight arrangements, visa processing, health certifications, manpower forecasts to overseas detachments, and disciplinary actions of 200+ aircrew candidates
- Practised crew resource management, Instrument Flight Rules, and tactical maneuvers in France and Australia
- Planned flight critical safety and survival re-training currencies as an Operations & Life Support Systems Officer

Projects and Extra Curricular Activities

NTU Overseas Entrepreneurship Programme

May 2023 – Dec. 2023

- Selected from a large pool of candidates to gain entrepreneurial experience with overseas start-ups
- Provided solutions to product and business issues while working closely with start-up founder
- Attended entrepreneurship workshops and provided detailed feedback and business plans for assigned start-up

Dyson-NTU Product Development Challenge

Jan. 2023 – May 2023

Project Title: Puffin - Cigarette Smoke Filtration Device

- Created a cigarette attachment to filter toxic particles, reducing health risks associated with secondhand smoke
- Collaborated with 3 team members to conduct market research through surveys, competitor analysis, focus groups and industry trend reports, using systems design methodologies to define our product vision
- Led iterative product prototyping via CAD, 3D printing, soldering of power delivery modules, & PM2.5 testing
- Presented our product to Dyson engineers & directors prior to James Dyson Award submission. Final grade: A-

Espressive Systems U Amaze Venture

Mar. 2023 – Apr. 2023

First Runners-Up

- Bested 11 teams in autonomously navigating a maze via obstacle avoidance with LiDAR while performing target alignment and collision using AprilTags for pose estimation
- Fabricated an autonomous mobile robot from scratch. Hardware was created using 3D printed parts, soldering, and breadboard circuitry. Software stack includes ROS, ESP32, and PID control on a differential drive system

NTU Students' Union Ministerial Open Discussions and Events Committee

Aug. 2022 – May 2023

Events Officer

- Spearheaded events geared towards open discussion forums between Singapore Ministers and NTU students
- Directed event proposals and cash flows towards logistics, catering, manpower, and building marketing materials
- Chief Facilitator for ForwardSG Youth Talk, moderating discussions on behalf of the National Youth Council

CERTIFICATIONS, SKILLS & INTERESTS

- **Certifications:** Nvidia Deep Learning Institute: Getting Started with Deep Learning
- **Programming Languages:** C, C++, Python, GLSL, SQL, Javascript
- **Tools:** ROS, Docker, Git, Notion, Linux, OpenCV, Open3D, PyTorch, Unity, Qt, Arduino, ESP, WebGL, [Three.js](#), [Node.js](#), React, Fusion360, MongoDB, Google Cloud, Microsoft Office, Davinci Resolve, Photoshop
- **Skills** Strategic planning, Leadership, Events management, Recruitment, Crew Resource Management
- **Languages:** Fluent in English and Mandarin, Basic conversational Thai
- **Interests** Tennis, Judo, Digital Art, Traveling, Photography